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V1 microcircuits underlying mouse visual behavior

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Visual behavior is based on the concerted activity of neurons in visual areas, where sensory signals are integrated with top-down information. In the past decade, the advent of new tools, such as functional imaging of populations of identified single neurons, high-density electrophysiology, virus-assisted circuit mapping, and precisely timed, cell-type specific manipulations, has advanced our understanding of the neuronal microcircuits underlying visual behavior. Studies in head-fixed mice, where such tools can routinely be applied, begin to provide new insights into the neural code of primary visual cortex (V1) underlying visual perception, and the micro-circuits of attention, predictive processing, and learning.

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Introduction

How do we see? This simple question does not have a simple answer. Vision is a complex process that starts with the absorption of light by the photoreceptors in the retina, which convey to downstream circuits the information necessary for perception and action. Essential for visual behavior are several processes, such as the encoding and transformation of visual representations across the visual hierarchy, the selection of salient or otherwise relevant information, the integration of visual inputs with learned priors or predictions for scene interpretation, and the translation into appropriate behavioral actions.

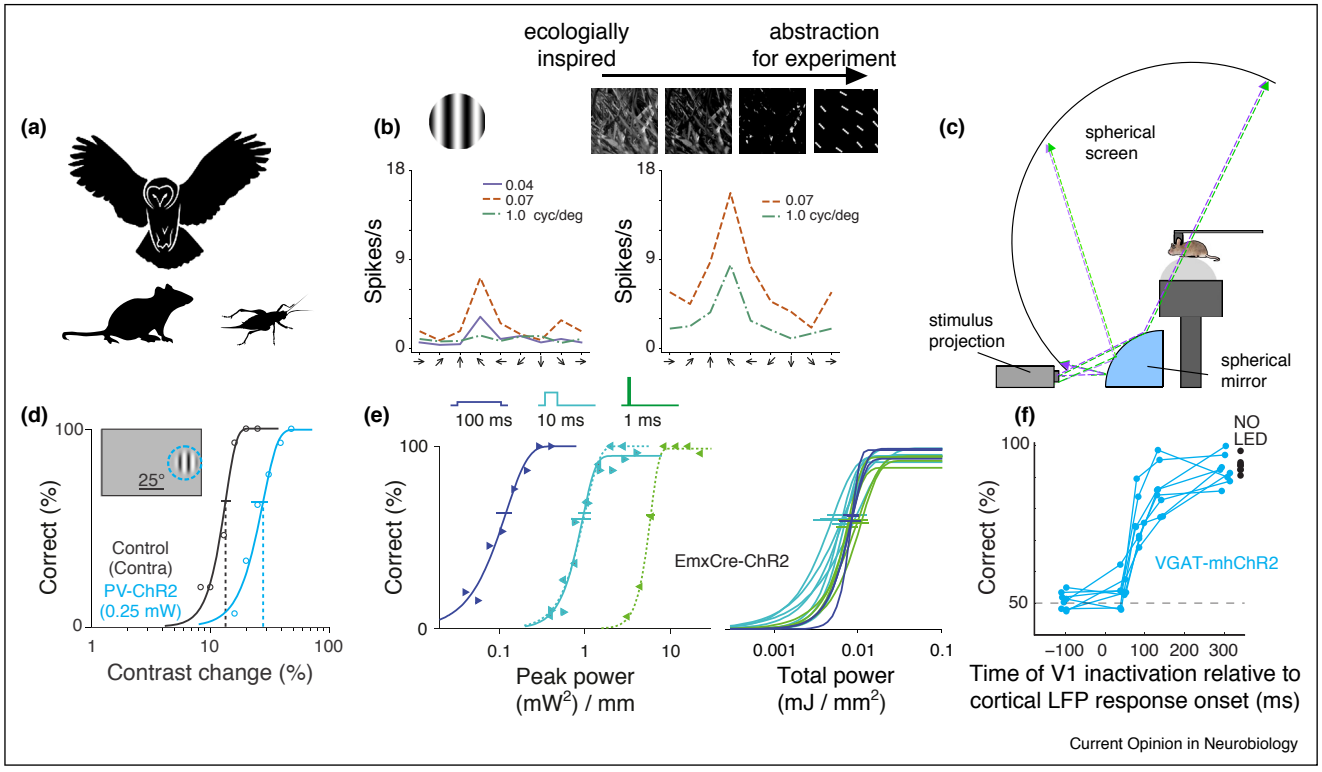
Within the last two decades, experiments in head-fixed mice have enabled researchers to investigate the neural basis of visual behavior on the microcircuit level. Rapid progress has been made, thanks to the availability of techniques that cannot as readily be used in other mammals. Examples are long-term monitoring of populations of identified single neurons using two-photon imaging with genetically encoded calcium indicators [1], and cell-type or projection-specific optogenetic manipulations [2]. Here, we survey this progress by focusing on *in vivo* studies in the head-fixed mouse, review how neural microcircuits in V1 support visual behavior and relate, where possible, experimental findings to computational models of visual function.

The mouse as a model system for visually guided behavior

While working with mice allows the interrogation of neuronal microcircuits at unprecedented detail, their suitability as a model system for visual behavior has often been called into question. Indeed, mice have large visual receptive fields (RFs, e.g. ~20 deg in V1 [3]) and poor spatial acuity in laboratory vision tests [4]. Yet, mice can discriminate not only simple, but also higher-order visual stimuli [5,6], as well as natural scenes [7]. Furthermore, they exhibit interesting and ecologically relevant visual behaviors: vision is their only sense to detect avian overhead predators [8,9], and is essential to orient themselves accurately towards prey and approach it from the distance [10] (Figure 1a).

It is becoming increasingly clear that considering the natural habitat of mice in the design and display of visual stimuli is likely to advance our understanding of mouse vision and visual behavior. For instance, preferences for spatial frequency in mouse V1 are concentrated below 0.08 cyc/deg, if measured with standard, artificial grating stimuli [11,12]. However, probing mouse V1 with more complex stimuli that mimic the optic flow experienced by running through grass (Figure 1b), revealed reliable, highly selective responses for orientation and direction, even at high spatial frequencies [13•]. Such adaptations in tuning might be understood in the context of efficient coding theory [14,15], which posits that perceptual systems have evolved towards transmission of as much information as possible, given the statistics of the natural environment and biological constraints of the nervous system. Relatedly, if performance in a natural task, such as object localization or identification, is taken as starting point, Bayesian ideal observer analysis [16] can provide a framework for extracting the most task-relevant stimulus features and hence infer representations for natural stimuli. Developing custom display solutions for optimal stimulation of the mouse's large field of view [12,17,18], and providing color inputs with wavelengths appropriate

Figure 1



Mouse vision and neural codes in V1. **(a)** Ethological visually guided behaviors in the mouse: avoidance of aerial predators [8,9] and prey capture [10]. **(b)** Comparison of V1 responses (example cell) to grating stimuli (left) and visual stimuli inspired by the mouse's visual habitat, here: patches of grass (right). These flow stimuli evoke higher firing rates and responses, which remain well tuned even for spatial frequencies so far considered beyond the resolution of the mouse (adapted from [13•]). **(c)** Custom projection and display devices to provide panoramic input of appropriate wavelength to the mouse's large field of view (Euler and Busse, unpublished). **(d)** Psychometric curve of an example mouse (single session) in a contrast change detection experiment, during control trials (black) and trials with V1 inactivation (blue) over V1 regions encoding the visual stimulus (inset). V1 inactivation systematically impairs perceptual performance, that is, shifts the psychometric curve to the right, providing evidence for a link between V1 activity and visual behavior (adapted from [30]). **(e)** Behavioral report of optogenetic V1 pyramidal cell stimulation can be well predicted by the total amount of light delivered. Example mouse, two sessions (adapted from [38•]). **(f)** During precisely timed, optogenetic inactivation of V1, behavioral performance in an orientation discrimination task is to large degree restored, even if as little as the initial ~80 ms of the stimulus-evoked V1 response is left intact (adapted from [34••]).

for their UV-sensitive cones [17–20] (Figure 1c), will likely reveal visual capabilities of mice that surpass what has so far been recognized [18], and will allow further testing of theories relating environment statistics to neural representations and task performance.

Do mice use the thalamocortical pathway for visual behavior?

Given the prominence of subcortical visual pathways in rodents, it has been imperative for studies of mouse visual behavior to demonstrate that, like in other mammals, specific visual tasks rely on thalamocortical pathways, in particular on V1. In the mouse, as much as 88% of retinal ganglion cells (RGCs) target the superior colliculus (SC) and most RGCs projecting to the dorsolateral geniculate nucleus of the thalamus (dLGN) have an axon collateral to SC [21]. In SC, neurons are selective for direction and orientation [22], and organized in maps with orientation columns [23,24]. Consistent with the importance of SC for visual behavior in rodents, ablation of primary visual cortex seems to have little impact on the detection and orientation towards salient visual stimuli [25,26]. Similarly, muscimol-induced inactivation of gerbil primary auditory cortex has little impact on behavioral performance in a sound detection task [27], and mice trained to detect objects using their whiskers can rapidly recover their performance level after ablation of barrel cortex [28], suggesting that across sensory systems rodents might not require primary cortices to detect simple stimuli. Remarkably, SC can have powerful control over visual cortex [29]: in postrhinal cortex, an extrastriate visual area, neural responses seem to remain intact after V1 inactivation, but they are abolished upon silencing SC. Despite the prominent role of the SC, short-term inactivation of V1 consistently demonstrated impairments of performance during orientation discrimination

[30–33,34••,35], motion perception [36], and detection of changes in stimulus contrast [30] (Figure 1d). Importantly, impairments in orientation discrimination performance seem to be strongest for low-contrast stimuli [32], and rather moderate suppression of firing rates can substantially elevate perceptual thresholds [30]. Together, these studies are consistent with a critical link between activity in mouse V1 and perceptual performance in a range of visual tasks.

Inactivating V1 in order to pinpoint its specific functions, however, comes with caveats. First, diminished behavioral performance during inactivation experiments could arise from effects unspecific to vision. A stronger case is thus made if V1 suppression impairs behavioral performance only in vision tasks, and not in a variant of the task in a different sensory modality [31]. Second, since V1 sends numerous projections to subcortical structures, manipulating V1 activity will inevitably also affect output targets, such as SC [37]. Hence, even if inactivation is targeted to area V1, it is challenging to ultimately rule out the involvement of SC in mouse visual perception.

The code and microcircuits for sensation and perception in mouse V1

While it seems that activity in mouse V1 is necessary for image-forming vision, it is less clear how many of its neurons are required for perception. This question has been elegantly tested in the somatosensory system's barrel cortex, where rats can report juxtacellular current injections into single neurons, leading to ~10–15 additional action potentials. Behavioral reports of such nanostimulation were most reliable if the stimulation targeted strongly responsive pyramidal cells in deep layers [39,40] or fast-spiking inhibitory interneurons [40,41]. Potentially related are findings of single 'hub' neurons in superficial layers of rodent V1, which exert widespread control over the cortical population: The induction of burst spiking in single neurons could change, in ~40% of the cases, global brain state as assessed by the local field potential (LFP) [42]. Induced bursts also affected, within a radius of ~100 μm, spiking activity in <1.7% of excitatory neurons [43,44] and in 29% of somatostatin-positive (SOM+) inhibitory interneurons [43]. The strong recruitment of inhibition resonates well with recent findings, where holographic optogenetic activation of a single pyramidal cell in L2/3 of mouse V1 induced a pattern of activity consistent with feature competition, in which similarly tuned neurons predominantly suppress each other [45••]. Together, these studies point towards an important role of V1 L2/3 circuits in suppressing redundant information and promoting sparse coding [46].

How activation of inhibitory interneurons in V1 influences visual perception is a matter of debate. One study reported that driving V1 parvalbumin-positive inhibitory interneurons (PV+) improved discriminability (d') for orientation [47]; however, the interpretation of these behavioral data is not straightforward, because they do not rule out the possibility that optogenetic stimulation affected task compliance [38•,48]. The opposite conclusions were reached by other studies, in which activation of PV+ interneurons impaired contrast discrimination, particularly in perceptually limited, threshold-level sensory conditions [48]. Interestingly, a similar decline in performance was observed for activation of somatostatin-positive (SOM+) interneurons, while activation of vasoactive intestinal peptide (VIP+) interneurons improved perceptual thresholds [48], potentially via the well-characterized disinhibitory VIP+ → SOM+ → pyramidal cell circuit [49]. An important next step will be a detailed characterization of visual perception under a wide range of optogenetic stimulation intensities. Such an approach would allow one to assess whether varying drive to the targeted interneurons changes perception in qualitatively different ways. This could potentially reconcile discrepancies between previous studies.

How relevant is spike timing in mouse V1 for driving visually guided behavior? A few studies causally tested, in the context of perception, predictions of theoretical models concerning the relevance of precise timing and synchronicity [50] versus the necessity of rate coding given chaotic network dynamics and noisy single neurons [51]. While single neuron nanostimulation in somatosensory cortex demonstrated that detection performance improved with the irregularity of the induced spike train [41], results in V1 obtained during larger-scale optogenetic stimulation of pyramidal cells rather suggest that the exact timing of spikes does not matter, at least in simple detection tasks [38•]. Indeed, detection performance of direct, low-intensity V1 optogenetic stimulation seems to depend only on total light energy, that is, the product of pulse intensity and duration, with little effect of stimulation frequency [38•] (Figure 1e). At least for the type of V1 stimulation employed, which increased spiking on average by ~1.1 spikes/s over 100 ms, behavior could be well approximated by a linear integration of spikes occurring within this stimulation window in a population of noisy neurons. These experiments elegantly and powerfully establish the capabilities of readout [52]; to which degree this result generalizes to more naturalistic stimulation patterns and is sufficient for explaining performance in complex, sensory tasks remain important open questions.

Which V1 spikes are critical for visual perception? Exploiting the power of precisely timed optogenetic suppression of V1 activity, Scanziani and colleagues could show that the first 40–80 ms of V1 activity are sufficient for mice to perform a simple orientation discrimination task (45 deg, [34••]) (Figure 1f). Remarkably, within this initial 80 ms of V1 response, the majority of V1 neurons

discriminating the stimulus had a response difference of ~1 spike. This highlights that the first V1 spikes, probably mostly representing the initial feedforward sweep, can reliably drive behavior, at least in simple discrimination tasks.

Finally, two photon functional imaging — currently still limited to the temporal scale of calcium indicators — allows to determine in large neuronal populations, which features of the neural code correlate with task performance. The heterogeneity of V1 L2/3 responses both before and during stimulus presentation, but not mean activity, seems a good candidate for predicting single trial detection performance and reaction time [53], and differs between active task performance and passive viewing [54•]. Other studies, however, have found that populations of V1 L2/3 neurons during orientation discrimination predominantly encoded stimulus identity as opposed to delayed behavioral choice [33], and did not alter selectivity after reward contingencies had been reversed [54•]. One goal for the future will be to investigate questions of population coding and behavior with more complex stimuli and tasks, at higher temporal resolution, on a trial-by-trial basis, and in individual animals [52]. Furthermore, our most detailed knowledge of the neural circuits underlying visual behavior in the mouse currently arises from layers 2/3 of primary visual cortex. In the future, it will be important to expand these investigations to other layers and areas beyond the primary visual cortex, whose potential function in mouse visual behavior remains largely unknown.

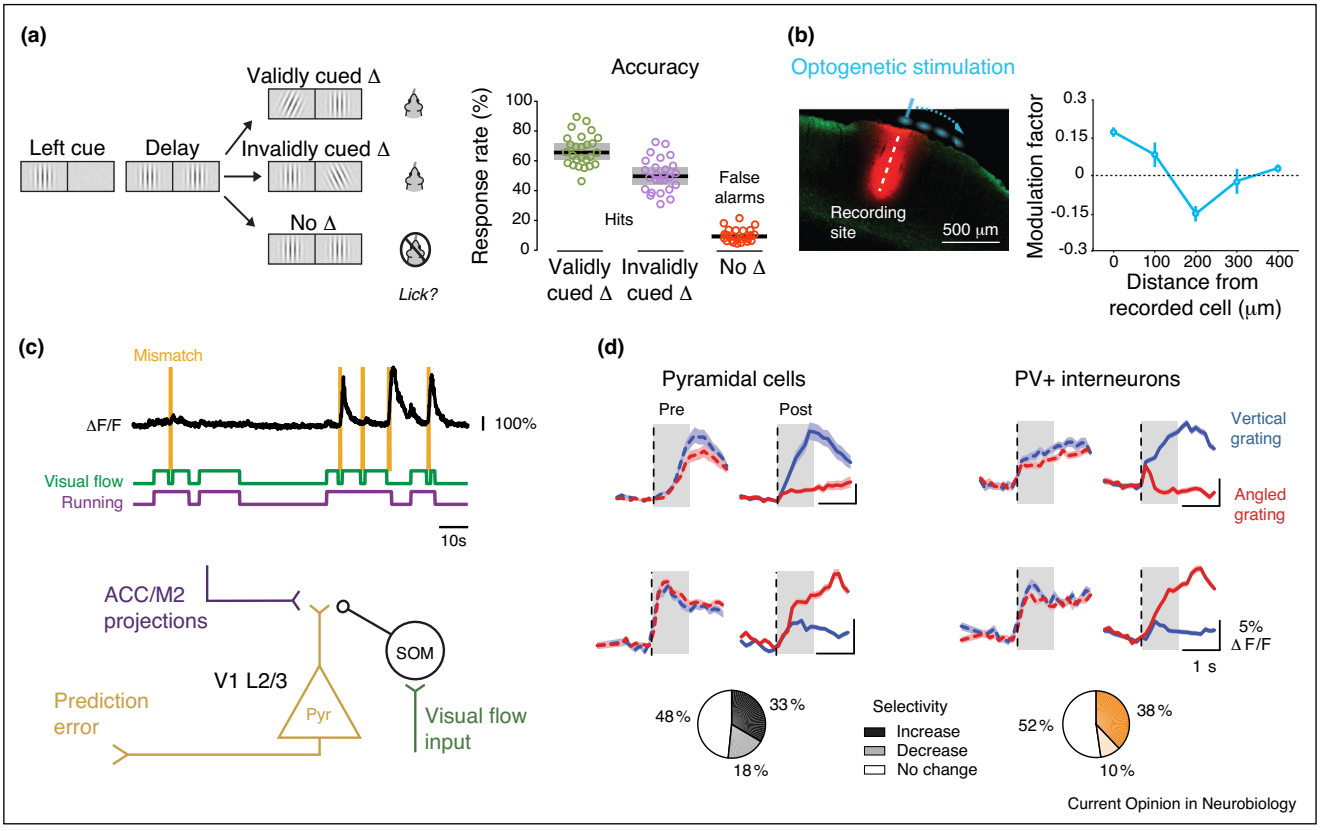
Top-down modulation of sensory processing in mouse V1

A fundamental task of sensory systems is to select behaviorally relevant information and enhance its neural representation, commonly referred to as selective attention. Selective visual attention has long been studied in primates, but recent work has shown that attention can also be probed in head-fixed mice [55•]. Indeed, in carefully designed spatial cueing tasks (Figure 2a), mice showed higher accuracy, faster reaction times, and enhanced perceptual sensitivity after valid compared to invalid cues [55•]; such results are characteristic signatures of selective visual attention. This key demonstration has opened up the possibility to relate perceptual improvements by attention to activity in V1 microcircuits, and to identify the neural source and signals controlling visual attention. Experiments on the microcircuits of attention would also allow testing predictions of a biophysically inspired computational model, which postulates specific roles for different types of interneurons in the attentional modulation of sensory responses [56].

One candidate source for attentional modulation of V1 responses is the anterior cingulate region of prefrontal cortex (ACC), which sends long-range, monosynaptic projections to V1 innervating pyramidal cells and all three major classes of inhibitory interneurons (PVs, SOMs, VIPs) [57,58]. In anesthetized mice, optogenetic stimulation of ACC projections terminating in V1 recruited these interneurons, whose concerted activity produced a pattern of pyramidal cell responses similar to the center-surround profile of spatial attention [58] (Figure 2b). It has also been reported that optogenetic stimulation of ACC improved behavioral performance during orientation discrimination [58], which is consistent with a functional relevance of those projections during task performance.

According to the influential framework of predictive coding, early processing stages are less concerned with selecting and representing stimulus features, but rather compare incoming sensory signals against an internal model of the world [59–61]. The predictive coding framework postulates a hierarchical processing scheme, where higher-level stages send predictions of sensory inputs to lower levels. Lower-level stages compare predicted against actual sensory input and might signal prediction errors, which are used to update the model. How these key components of the predictive coding framework might be implemented by neural circuits in mouse cortex has been tested in a series of studies, in which mice were navigating virtual reality (VR) environments [57,62,63,64••,65]. VR environments allow sudden perturbations of the optic flow, creating visual input that deviates from what would be expected based on self-motion. Two-photon calcium imaging revealed that such mismatches evoked responses in a population of layer 2/3 V1 neurons [62,65] (Figure 2c), consistent with prediction error signals. Such mismatch signals required that the animals experienced the coupling between optic flow and self-motion, because mismatch responses were absent in a separate group of animals receiving identical visual stimulation without visuomotor coupling [64••]. The predictions themselves are thought to be conveyed by subregions of the ACC and secondary motor cortex (M2). ACC/M2 provide, via topographically organized, monosynaptic projections, to V1 layer 2/3 neurons motor-related signals [57]. Motor-related signals in ACC/M2 projections were found to be correlated with the optic flow expected by self-movement, and this correlation was restored, after additional practice, in VR environments, where the coupling of optic flow to the animal's movement direction was inverted [57]. Finally, a circuit in layer 2/3 of mouse V1 has been proposed that could potentially implement the computation of prediction errors (Figure 2c). Layer 2/3 'mismatch' neurons integrate motor-related, excitatory input and inhibitory input from SOM+ interneurons driven by the visual optic flow. Without perturbations, the excitatory motor-input and the inhibitory visual input cancel out; if perturbations stop the optic flow, however, SOM+ activity decreases and this release of inhibition allows the mismatch neuron to signal a prediction error [64••].

Figure 2



Top-down modulations of V1 circuits. **(a)** The adaptation of classic spatial cueing tasks to head-fixed mice opens up the possibility to study microcircuits of attentional selection. Left: in the experimental setup, the animal is facing two screens, one of which displays a cue grating at the beginning of the trial. After the following delay period, where the grating is presented on both screens, the animal needs to lick if either of the two gratings changes its orientation and refrain from licking if the orientations remain the same. Right: behavioral performance is better in trials in which the cue is valid (i.e. presented at the same location at which a later change in orientation needs to be detected) compared to trials in which the cue is invalid (adapted from [55•]). **(b)** In anesthetized mice, optogenetic activation of ACC projections over V1 results in center-surround modulation. Left: laser stimulation sites (blue) relative to the recording site (red). Right: Spatial profile of response modulation, averaged across the population of recorded neurons. A facilitatory center and a suppressive surround are reminiscent of spatial attention effects (adapted from [58]). **(c)** A predicive coding interpretation of V1 responses. Top: Sudden halts (yellow) of optic flow in a VR environment can elicit 'mismatch' responses from a population of L2/3 V1 pyramidal cells. Bottom: The proposed microcircuit generating these responses integrates excitatory signals from ACC/M2 (purple) signaling predicted optic flow and inhibitory inputs from SOM+ neurons (orange) driven by the visual optic flow (green). If visual and predicted optic flow match, these inputs cancel; in case of 'mismatches' the pyramidal cell produces a prediction error (adapted from [64••]). **(d)** Top: During orientation discrimination learning, individual pyramidal cells and PV+ interneurons develop a preference for one or the other grating stimuli. Bottom: Population summary of changes in selectivity for these two neuron types (adapted from [67••]).

VR experiments with rodents also highlight the intimate relation between vision and navigation. As mice navigate a virtual corridor lined by visual stimuli, responses of V1 neurons to identical stimuli were found to be selective for the animal's spatial position in the corridor [66••,63]. V1 selectivity for spatial position closely corresponded to the spatial position encoded in the hippocampal area CA1, and population activity in both regions seemed to reflect the animal's subjective estimate of position [66••].

Key ideas behind theories of selective attention or predictive processing already imply that neural representations in V1 are not static, but depend on context and experience. How experience, or learning, affects sensory processing in V1 circuits can be investigated elegantly in the mouse, where the activity of populations of identified single neurons can be tracked across recording sessions. Longitudinal two-photon calcium imaging during learning of a visual detection task, for instance, revealed decreasing responses of layer 4 neurons, the primary target of bottom-up inputs, yet increasing activity in top-down projections from the retrosplineal cortex [68]. Consistent with such learning-related strengthening of top-down influences on V1 processing, two studies have reported that L2/3 neurons can acquire ramp-like responses, possibly encoding the timing of the behaviorally relevant stimulus [31,68].

Besides strengthening top-down signals, learning can also improve V1 stimulus processing [31,32]. Indeed, learning to discriminate stimulus orientation made responses more reliable in neurons that already preferred either orientation before learning; other neurons developed such relative preference during learning [31] (Figure 2d). On the microcircuit-level, a methodological tour de force combining longitudinal functional imaging with cell type identification by post-mortem immunohistochemistry [67••] could reveal differential effects of learning on multiple classes of inhibitory interneurons. With learning, PV+ neurons acquired substantial stimulus selectivity (Figure 2d), which could be modeled as increased functional connectivity between pyramidal and PV+ cells with similar preferences. The model also revealed that the magnitude of learning-related increases in stimulus preference of pyramidal cells was inversely related to their pre-learning coupling strength with SOM+ neurons. Thus, SOM+ neurons might be responsible for gating learning-induced effects in V1. Consistent with a role in gating learning-induced changes, the activity of SOM+ interneurons was found to be reduced during learning [68], and rather decorrelated from the local network [67••]. Thus, a release from inhibition during learning might be important to promote plasticity in V1 [68], where optogenetic activation of SOM+ neurons could, to some degree, reverse learning-induced changes [68].

Although learning-related improvements in behavioral performance are often paralleled by changes in sensory processing, the time course for behavioral and neural measures of learning can also dissociate. During classical conditioning of orientation discrimination, for instance, sensory processing in mouse V1 improved before successful discrimination was evident in the animal's licking behavior [32]. This observation might be related to reports from the auditory system, showing elegantly that behavioral readouts of learning can systematically underestimate the acquisition of task knowledge [69]. These findings provide important constraints for theories of learning, and highlight the need to quantify learning precisely in individual animals, and to carefully choose behavioral indicators of progress.

Conclusions and outlook

We reviewed recent work on how microcircuits in V1, and projections from top-down sources, might serve visual behavior of the mouse. While rapid progress has been made in understanding the role of V1 upper layers and their specific inputs, much less is understood about the role of lower-layer circuits and inter-laminar connectivity. A new wave of discoveries is likely to come from recordings with high-density CMOS-based electrodes [70], sampling from distributed areas of the brain [71], or from brain-wide functional imaging methods [72], which promise to frame V1 computations in the brain-wide network of regions active during visual behavior.

Conflict of interest statement

Nothing declared.

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References and recommended reading

Papers of particular interest, published within the period of review, have been highlighted as:

- • of special interest
- •• of outstanding interest

1. Lin MZ, Schnitzer MJ: **Genetically encoded indicators of neuronal activity**. *Nat Neurosci* 2016, **19**:1142-1153.
2. Kim CK, Adhikari A, Deisseroth K: **Integration of optogenetics with complementary methodologies in systems neuroscience**. *Nat Rev Neurosci* 2017, **18**:222-235.
3. Niell CM, Stryker MP: **Highly selective receptive fields in mouse visual cortex**. *J Neurosci* 2008, **28**:7520-7536.
4. Prusky GT, West PWR, Douglas RM: **Behavioral assessment of visual acuity in mice and rats**. *Vis Res* 2000, **40**:2201-2209.
5. Khastkhodaei Z, Jurjut O, Katzner S, Busse L: **Mice can use second-order, contrast-modulated stimuli to guide visual perception**. *J Neurosci* 2016, **36**:4457-4469.
6. Schnabel UH, Bossens C, Lorteije JAM, Self MW, Beeck de HO, Roelfsema PR: **Figure-ground perception in the awake mouse and neuronal activity elicited by figure-ground stimuli in primary visual cortex**. *Sci Rep* 2018, **8**:1-14.
7. Yu Y, Hira R, Stirman JN, Yu W, Smith IT, Smith SL: **Mice use robust and common strategies to discriminate natural scenes**. *Sci Rep* 2018, **8**:1379.
8. Yilmaz M, Meister M: **Rapid innate defensive responses of mice to looming visual stimuli**. *Curr Biol* 2013, **23**:2011-2015.
9. De Franceschi G, Vivattanasarn T, Saleem AB, Solomon SG: **Vision guides selection of freeze or flight defense strategies in mice**. *Curr Biol* 2016, **26**:2150-2154.
10. Hoy JL, Yavorska I, Wehr M, Niell CM: **Vision drives accurate approach behavior during prey capture in laboratory mice**. *Curr Biol* 2016, **26**:3046-3052.
11. Gao E, DeAngelis GC, Burkhalter A: **Parallel input channels to mouse primary visual cortex**. *J Neurosci* 2010, **30**:5912-5926.
12. Marshel JH, Garrett ME, Nauhaus I, Callaway EM: **Functional specialization of seven mouse visual cortical areas**. *Neuron* 2011, **72**:1040-1054.
13. • Dyballa L, Hoseini MS, Dadarlat MC, Zucker SW, Stryker MP: **Flow stimuli reveal ecologically appropriate responses in mouse visual cortex**. *Proc Natl Acad Sci U S A* 2018, **115**:11304-11309.
Using novel visual stimuli inspired by the optic flow likely experienced by the mouse while running through patchy grass, this study demonstrates that neurons in mouse V1 can be responsive and selective at spatial frequencies higher than previously appreciated.
14. Barlow H: **Possible principles underlying the transformations of sensory messages — MIT Press Scholarship**. *Sensory Communication*. Cambridge, MA: MIT Press; 1961, 217-234.
15. Attneave F: **Some informational aspects of visual perception**. *Psychol Rev* 1954, **61**:183-193.
16. Geisler WS: **Visual perception and the statistical properties of natural scenes**. *Annu Rev Psychol* 2008, **59**:167-192.
17. Rhim I, Coello-Reyes G, Ko H, Nauhaus kI: **Maps of cone opsin input to mouse V1 and higher visual areas**. *J Neurophysiol* 2017. jn.00849.2016.
18. Denman DJ, Luviano JA, Ollerenshaw DR, Cross S, Williams D, Buice MA, Olsen SR, Reid RC: **Mouse color and wavelength-**

specific luminance contrast sensitivity are non-uniform across visual space. *eLife* 2018, **7**:e31209.

19. Tan Z, Sun W, Chen TW, Kim D, Ji N: **Neuronal representation of ultraviolet visual stimuli in mouse primary visual cortex**. *Sci Rep* 2015, **5**:12597.

20. Denman DJ, Siegle JH, Koch C, Reid RC, Blanche TJ: **Spatial organization of chromatic pathways in the mouse dorsal lateral geniculate nucleus**. *J Neurosci: Off J Soc Neurosci* 2017, **37**:1102-1116.

21. Ellis EM, Gauvain G, Sivyer B, Murphy GJ: **Shared and distinct retinal input to the mouse superior colliculus and dorsal lateral geniculate nucleus**. *J Neurophysiol* 2016, **116**:602-610.

22. Wang L, Sarnaik R, Rangarajan K, Liu X, Cang J: **Visual receptive field properties of neurons in the superficial superior colliculus of the mouse**. *J Neurosci* 2010, **30**:16573-16584.

23. Ahmadlou M, Heimel JA: **Preference for concentric orientations in the mouse superior colliculus**. *Nat Commun* 2015, **6**: ncomms7773.

24. Feinberg EH, Meister M: **Orientation columns in the mouse superior colliculus**. *Nature* 2015, **519**:229-232.

25. Schneider GE: **Two visual systems**. *Science* 1969, **163**:895-902.

26. Petruno SK, Clark RE, Reinagel P: **Evidence that primary visual cortex is required for image, orientation, and motion discrimination by rats**. *PLOS ONE* 2013, **8**:e56543.

27. Caras ML, Sanes DH: **Top-down modulation of sensory cortex gates perceptual learning**. *Proc Natl Acad Sci U S A* 2017, **114**:9972-9977.

28. Hong YK, Lacefield CO, Rodgers CC, Bruno RM: **Sensation, movement and learning in the absence of barrel cortex**. *Nature* 2018, **561**:542-546.

29. Beltramo R, Scanziani M: **A collicular visual cortex: neocortical space for an ancient midbrain visual structure**. *Science* 2019, **363**:64-69.

30. Glickfeld LL, Histed MH, Maunsell JHR: **Mouse primary visual cortex is used to detect both orientation and contrast changes**. *J Neurosci* 2013, **33**:19416-19422.

31. Poort J, Khan AG, Pachitariu M, Nemri A, Orsolic I, Krupic J, Bauza M, Sahani M, Keller GB, Mrsic-Flogel TD *et al.*: **Learning enhances sensory and multiple non-sensory representations in primary visual cortex**. *Neuron* 2015, **86**:1478-1490.

32. Jurjut O, Georgieva P, Busse L, Katzner S: **Learning enhances sensory processing in mouse V1 before improving behavior**. *J Neurosci* 2017, **37**:6460-6474.

33. Goard MJ, Pho GN, Woodson J, Sur M: **Distinct roles of visual, parietal, and frontal motor cortices in memory-guided sensorimotor decisions**. *eLife* 2016, **5**:e13764.

34. •• Resulaj A, Ruediger S, Olsen SR, Scanziani M: **First spikes in visual cortex enable perceptual discrimination**. *eLife* 2018, **7**: e34044.
Using precisely timed optogenetic inactivation of primary visual cortex, this study demonstrates that the first stimulus-evoked spikes can be sufficient for orientation discrimination.

35. Zatka-Haas P, Steinmetz NA, Carandini M, Harris KD: **Distinct contributions of mouse cortical areas to visual discrimination**. *bioRxiv* 2019:501627.

36. Marques T, Summers MT, Fioreze G, Fridman M, Dias RF, Feller MB, Petreanu L: **A role for mouse primary visual cortex in motion perception**. *Curr Biol* 2018, **28** 1703–1713.e6.

37. Ahmadlou M, Tafreshiha A, Heimel JA: **Visual cortex limits pop-out in the superior colliculus of awake mice**. *Cereb Cortex* 2017, **27**:5772-5783.

38. • Histed MH, Maunsell JHR: **Cortical neural populations can guide behavior by integrating inputs linearly, independent of synchrony**. *Proc Natl Acad Sci U S A* 2014, **111**:E178-E187.
This study tested the ability to detect direct, optogenetic activation of V1 excitatory neurons and found that detection performance was driven by the total amount of light delivered, irrespective of the exact pattern or frequency of stimulation. V1 electrophysiological recordings showed that the light-induced changes in firing rates were small, and behavior reflected the integration of the total number of spikes over the 100 ms of stimulation.

39. Houweling AR, Brecht M: **Behavioural report of single neuron stimulation in somatosensory cortex**. *Nature* 2008, **451**:65-68.

40. Tanke N, Borst JGG, Houweling AR: **Single-cell stimulation in barrel cortex influences psychophysical detection performance**. *J Neurosci* 2018, **38**:2057-2068.

41. Doron G, von Heimendahl M, Schlattmann P, Houweling AR, Brecht M: **Spiking irregularity and frequency modulate the behavioral report of single-neuron stimulation**. *Neuron* 2014, **81**:653-663.

42. Li CyT, Poo M, Dan mY: **Burst spiking of a single cortical neuron modifies global brain state**. *Science* 2009, **324**:643-646.

43. Kwan AC, Dan Y: **Dissection of cortical microcircuits by single-neuron stimulation in vivo**. *Curr Biol* 2012, **22**:1459-1467.

44. Meyer JF, Golshani P, Smirnakis SM: **The effect of single pyramidal neuron firing within layer 2/3 and layer 4 in mouse V1**. *Front Neural Circuits* 2018:12.

45. •• Chettih SN, Harvey CD: **Single-neuron perturbations reveal feature-specific competition in V1**. *Nature* 2019, **567**:334.
Using holographic optogenetic stimulation of single V1 L2/3 pyramidal cells the authors demonstrate suppression between similarly tuned neurons, pointing towards feature competition and sparse coding.

46. Olshausen BA, Field DJ: **Emergence of simple-cell receptive field properties by learning a sparse code for natural images**. *Nature* 1996, **381**:607-609.

47. Lee SH, Kwan AC, Zhang S, Phoumthipphavong V, Flannery JG, Masmanidis SC, Taniguchi H, Huang ZJ, Zhang F, Boyden ES *et al.*: **Activation of specific interneurons improves V1 feature selectivity and visual perception**. *Nature* 2012, **488**:379-383.

48. Cone JJ, Scantlen MD, Histed MH, Maunsell JHR: **Different Inhibitory Interneuron Cell Classes Make Distinct Contributions to Visual Contrast Perception**. *ENeuro* 2019:6. ENEURO.0337-18.2019.

49. Pfeffer CK, Xue M, He M, Huang ZJ, Scanziani M: **Inhibition of inhibition in visual cortex: the logic of connections between molecularly distinct interneurons**. *Nat Neurosci* 2013, **16**:1068-1076.

50. Fries P: **Rhythms for cognition: communication through coherence**. *Neuron* 2015, **88**:220-235.

51. van Vreeswijk C, Sompolinsky H: **Chaos in neuronal networks with balanced excitatory and inhibitory activity**. *Science* 1996, **274**:1724-1726.

52. Panzeri S, Harvey CD, Piasini E, Latham PE, Fellin T: **Cracking the neural code for sensory perception by combining statistics, intervention, and behavior**. *Neuron* 2017, **93**:491-507.

53. Montijn JS, Goltstein PM, Pennartz CM: **Mouse V1 population correlates of visual detection rely on heterogeneity within neuronal response patterns**. *eLife* 2015:4.

54. • Pho GN, Goard MJ, Woodson J, Crawford B, Sur M: **Task-dependent representations of stimulus and choice in mouse parietal cortex**. *Nat Commun* 2018, **9**:2596.
This study compares V1 and PPC neurons in terms of selectivity for visual input and choice signals. While activity V1 is largely driven by visual stimuli and rather insensitive to the mapping between stimuli and choice, PPC activity is task-dependent and strongly correlated with behavioral choices.

55. • Wang L, Krauzlis RJ: **Visual selective attention in mice**. *Curr Biol* 2018, **28** 676–685.e4..
The authors develop several variants of a spatial cueing task for head-fixed mice, in which behavior shows hallmarks of attentional selection. This development opens up the possibility to study, in the mouse, microcircuits of selective visual attention.

56. Buia CI, Tiesinga PH: **Role of interneuron diversity in the cortical microcircuit for attention**. *J Neurophysiol* 2008, **99**:2158-2182.

57. Leinweber M, Ward DR, Sobczak JM, Attinger A, Keller GB: **A sensorimotor circuit in mouse cortex for visual flow predictions**. *Neuron* 2017, **95** 1420–1432.e5.

58. Zhang S, Xu M, Kamigaki T, Hoang Do JP, Chang WC, Jenvay S, Miyamichi K, Luo L, Dan Y: **Selective attention. Long-range and local circuits for top-down modulation of visual cortex processing**. *Science (New York, N.Y.)* 2014, **345**:660-665.

59. Rao RPN, Ballard DH: **Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects**. *Nat Neurosci* 1999, **2**:79-87.

60. Spratling MW: **Predictive coding as a model of response properties in cortical area V1**. *J Neurosci* 2010, **30**:3531-3543.

61. Keller GB, Mrsic-Flogel TD: **Predictive processing: a canonical cortical computation**. *Neuron* 2018, **100**:424-435.

62. Keller GB, Bonhoeffer T, Hübener M: **Sensorimotor mismatch signals in primary visual cortex of the behaving mouse**. *Neuron* 2012, **74**:809-815.

63. Fiser A, Mahringer D, Oyibo HK, Petersen AV, Leinweber M, Keller GB: **Experience-dependent spatial expectations in mouse visual cortex**. *Nat Neurosci* 2016, **19**:1658-1664.

64. •• Attinger A, Wang B, Keller GB: **Visuomotor coupling shapes the functional development of mouse visual cortex**. *Cell* 2017, **169** 1291–1302.e14.
This study demonstrates that responses to sensorimotor mismatch in V1 L2/3 neurons require experience with normal sensorimotor coupling. In addition, the study proposes a microcircuit for computation of prediction error signals consisting of convergent inputs to 'mismatch neurons' of motor-related inputs and inhibitory inputs from SOM+ neurons, encoding the sensory stimulus. Under normal conditions, these inputs cancel; during sudden halts of the visual optic flow, the 'mismatch' neuron is released from inhibition and signals a mismatch response.

65. Zmarz P, Keller GB: **Mismatch receptive fields in mouse visual cortex**. *Neuron* 2016, **92**:766-772.

66. •• Saleem AB, Diamanti EM, Fournier J, Harris KD, Carandini M: **Coherent encoding of subjective spatial position in visual cortex and hippocampus**. *Nature* 2018, **562**:124-127.
This study highlights the intimate relation between vision and navigation. It shows that identical visual stimuli lining the walls of a virtual corridor at different spatial positions elicit differential responses in neurons of mouse V1. Furthermore, not only hippocampal representations but also those in V1 seem to reflect subjective estimates of the mouse's spatial position.

67. •• Khan AG, Poort J, Chadwick A, Blot A, Sahani M, Mrsic-Flogel TD, Hofer SB: **Distinct learning-induced changes in stimulus selectivity and interactions of GABAergic interneuron classes in visual cortex**. *Nat Neurosci* 2018, **21**:851-859.
Combining longitudinal two-photon calcium imaging of populations of single neurons with post-mortem immunohistochemical identification of cell types, this study demonstrates that learning differentially shapes stimulus selectivity and functional connectivity of L2/3 pyramidal neurons, and PV+, SOM+, and VIP+ inhibitory interneurons. During learning, both pyramidal and PV+ cells increase their stimulus preference and their preference-specific functional coupling. Pre-learning coupling of SOM+ neurons to pyramidal cells was found to be inversely related to learning-related increases in preferences, suggesting that SOM+ might gate learning.

68. Makino H, Komiyama T: **Learning enhances the relative impact of top-down processing in the visual cortex**. *Nat Neurosci* 2015, **18**:1116-1122.

69. Kuchibhotla KV, Sten TH, Papadoyannis ES, Elnozahy S, Fogelson KA, Kumar R, Boubenec Y, Holland PC, Ostojic S, Froemke RC: **Dissociating task acquisition from expression during learning reveals latent knowledge**. *Nat Commun* 2019, **10**:1-13.

70. Jun JJ, Steinmetz NA, Siegle JH, Denman DJ, Bauza M, Barbarits B, Lee AK, Anastassiou CA, Andrei A, Aydn Ç *et al.*: **Fully integrated silicon probes for high-density recording of neural activity**. *Nature* 2017, **551**:24636.

71. Steinmetz N, Zatka-Haas P, Carandini M, Harris K: **Distributed correlates of visually-guided behavior across the mouse brain**. *bioRxiv* 2018:474437.

72. Macé E, Montaldo G, Trenholm S, Cowan C, Brignall A, Urban A, Roska B: **Whole-brain functional ultrasound imaging reveals brain modules for visuomotor integration**. *Neuron* 2018, **100** 1241–1251.e7.